

Exercise I. [course]

[12 marks]

Let $\alpha, \beta \in \mathbb{R}$. Show that

⑥ 1. $I(\alpha, \beta) = \int_e^\infty \frac{1}{x^\alpha \ln^\beta x} dx$ converges $\Leftrightarrow (\alpha > 1)$ or $(\alpha = 1 \text{ and } \beta > 1)$.

⑥ 2. $J(\alpha, \beta) = \int_0^{\frac{1}{2}} \frac{1}{x^\alpha |\ln x|^\beta} dx$ converges $\Leftrightarrow (\alpha < 1)$ or $(\alpha = 1 \text{ and } \beta > 1)$.

Exercise II.

[15 marks]

Study the nature of the following improper integrals :

③ x 5 1. $\int_0^{+\infty} \frac{\sin x}{1 + \cos x + e^x} dx,$ 3. $\int_0^\pi \frac{x - \frac{\pi}{2}}{\cos x \sqrt{\sin x}} dx,$ 5. $\int_1^{+\infty} \frac{|\cos x|}{x^4} dx.$
2. $\int_0^\infty \frac{\sqrt{x} \sin(\frac{1}{x})}{\ln(1+x)} dx,$ 4. $\int_1^{+\infty} \frac{\cos x}{x^4} dx,$

Exercise III.

[6 marks]

Determine the following limits :

③ 1. $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{(2n+5k)^2}{n^2},$

③ 2. $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n (\ln(5k+n) - \ln n).$

Exercise IV.

[13 marks]

Study the nature of the following series :

② 1. $\sum_{n=1}^\infty \frac{1}{2^n},$ ② 3. $\sum_{n=1}^\infty \sin(\frac{n!}{n^n}),$ ③ 5. $\sum_{n=1}^\infty \frac{(-1)^n + 1}{\sqrt{n}}.$
③ 2. $\sum_{n=1}^\infty \frac{n!}{n^n},$ ③ 4. $\sum_{n=1}^\infty \frac{1}{\ln n} \sin(\frac{2\pi n}{3}),$

Hint : You may use part 2. to study the nature of the series in part 3..

Exercise V.

[12 marks]

Study, according to the values of $\alpha \in \mathbb{R}$, the nature of the following series and integrals :

③ 1. $\int_0^{+\infty} \frac{\arctan x}{x^\alpha} dx$ ③ 3. $\sum_{n=1}^\infty \ln(1+n^\alpha).$
③ 2. $\int_\alpha^{+\infty} \frac{1}{(x-2)\sqrt{|x-3|}} dx,$ ③ 4. $\sum_{n=1}^\infty \frac{\alpha^n}{n!}.$

Exercise VI.

[8 marks]

Let $n \in \mathbb{N}$ and f be a continuous function near zero such that $f(t) = o_{t \rightarrow 0}(t^n)$, that is, $\frac{f(t)}{t^n} \xrightarrow{t \rightarrow 0} 0$.

④ 1. Let $F(x) = \int_0^x f(t) dt$. Using the mean value theorem for integrals, show that $\lim_{x \rightarrow 0} \frac{|F(x)|}{|x^{n+1}|} = 0$.

② 2. Deduce that $F(x) = o_{x \rightarrow 0}(x^{n+1})$.

② 3. Assume that the Maclurin's expansion of

$$\frac{1}{1+t^2} = 1 - t^2 + t^4 - \dots + (-1)^m t^{2m} + o_{t \rightarrow 0}(t^{2m+1}).$$

Deduce the the Maclurin's expansion of $\arctan x$.

Exercise VII.

[4 marks]

④ Let $f : [0, 1] \mapsto \mathbb{R}$ be a continuous function and $a_n = \frac{1}{n} \int_0^1 t^n f(t) dt$. Show that the series $\sum_{n=1}^\infty a_n$ is absolutely convergent.

①

ExI. Let $\alpha, \beta \in \mathbb{R}$. Show that

1. $I(\alpha, \beta) = \int_e^\infty \frac{1}{x^\alpha \ln^\beta x} dx$ converges $\Leftrightarrow (\alpha > 1) \text{ or } (\alpha = 1 \text{ and } \beta > 1)$.

2. $J(\alpha, \beta) = \int_0^1 \frac{1}{x^\alpha |\ln x|^\beta} dx$ converges $\Leftrightarrow (\alpha < 1) \text{ or } (\alpha = 1 \text{ and } \beta > 1)$
sol:

1. we discuss three cases

• $\alpha > 1$: choose $\mu \in]1, \alpha[$, for ex $\mu = \frac{1+\alpha}{2}$, since $\lim_{x \rightarrow \infty} \frac{x^\mu}{x^\alpha \ln^\beta x} = 0$ and $\mu > 1$, then $I(\alpha, \beta)$ converges.

• $\alpha < 1$: choose $\mu \in]\alpha, 1[$ for ex $\mu = \frac{1+\alpha}{2}$, since $\lim_{x \rightarrow \infty} \frac{x^\mu}{x^\alpha \ln^\beta x} = \infty$ and $\mu < 1$, then $I(\alpha, \beta)$ diverges.

• $\alpha = 1$: we perform the change of variable $x = u(t) = e^t$ which is of class C^1 and st $\uparrow$ on $[1, \infty[$. Then $\int_e^\infty \frac{1}{x \ln^\beta x} dx = \int_1^\infty \frac{1}{t^\beta} dt$ Converges if $\beta > 1$. ~~Then we conclude that~~

Thus we conclude that $I(\alpha, \beta)$ converges iff $(\alpha > 1) \text{ or } (\alpha = 1 \text{ and } \beta > 1)$.

2. We perform the change of variable $x = u(t) = \frac{1}{e^t}$ which is of class C^1 and st $\downarrow$ on $[e, \infty[$. Then $J(\alpha, \beta) =$ Then

$$J(\alpha, \beta) = \int_0^1 \frac{1}{x^\alpha |\ln x|^\beta} dx = \int_1^{+\infty} \frac{1}{t^{-\alpha+2} \ln^\beta t} dt. \quad \text{Hence by part 1.}$$

$$J(\alpha, \beta) \text{ conv} \Leftrightarrow (-\alpha+2 > 1) \text{ or } (-\alpha+2 = 1 \text{ and } \beta > 1) \Leftrightarrow (\alpha < 1) \text{ or } (\alpha = 1 \text{ and } \beta > 1)$$

②

①

Ex 8.

Study the nature of the following imp integrals:

$$1. I_1 = \int_0^{+\infty} \frac{\sin x}{1 + \cos x + e^x} dx.$$

$$\textcircled{1/1} \text{ Near } +\infty: \left| \frac{\sin x}{1 + \cos x + e^x} \right| = \frac{|\sin x|}{\underbrace{1 + \cos x + e^x}_{\geq e^x}} \leq \frac{1}{e^x}$$

$$\textcircled{1/2} x^{100} \cdot \frac{1}{e^x} \xrightarrow{x \rightarrow \infty} 0 \Rightarrow \int_0^{+\infty} \frac{1}{e^x} dx \text{ converges} \Rightarrow I_1 \text{ is absol conv (comparison test)}$$

$$2. I_2 = \int_0^{\infty} \frac{\sqrt{x} \sin(\frac{1}{x^2})}{\ln(1+x)} dx$$

$$\text{Near } 0^+: |f(x)| \stackrel{\textcircled{1/1}}{=} \frac{\sqrt{x}}{\ln(1+x)} \sim \frac{\sqrt{x}}{x} = \frac{1}{x^{1/2}} \quad (\alpha = 1/2 < 1, \text{ absol conv.})$$

$$\text{Near } \infty: 0 \leq f(x) \stackrel{\textcircled{1/1}}{\sim} \frac{\sqrt{x} \cdot \frac{1}{x^2}}{\ln(1+x)} = \frac{1}{x^{3/2} \ln(1+x)} \stackrel{\textcircled{1/1}}{<} \frac{1}{x^{3/2}} \quad (\alpha = 3/2 > 1, \text{ conv by comp \& equiv test})$$

Thus I_2 converges.

$$\frac{1}{x^{3/2} \ln(1+x)} \xrightarrow{x \rightarrow +\infty} 0 \text{ absol conv.}$$

$$3. I_3 = \int_0^{\pi} \frac{x - \frac{\pi}{2}}{\cos x \sqrt{\sin x}} dx$$

$$\textcircled{1} \text{ Near } 0^+: f(x) \sim \frac{-\pi/2}{1 - x^{1/2}} < 0 \text{ convergent}$$

$$\textcircled{1} \text{ Near } \pi/2: \frac{x - \pi/2}{\cos x \sqrt{\sin x}} \xrightarrow{x \rightarrow \pi/2} -1 \text{ no singular pt (conv)}$$

$$\textcircled{1} \text{ Near } \pi^-: \frac{x - \pi/2}{\cos x \sqrt{\sin x}} \sim \frac{\pi/2}{-1 \sqrt{(\pi - x)(-1)}} = \frac{-\pi/2}{\sqrt{(\pi - x)^{1/2}}} \quad (\text{conv})$$

$\uparrow$ Taylor

Thus I_3 is convergent.

EX II

$$4. I_4 = \int_1^{+\infty} \frac{\cos x}{x^{1/4}} dx$$

nearby: If we perform integration by parts, we obtain

$$I_4 = \int_1^{+\infty} \frac{1}{x^{1/4}} \underbrace{\cos x}_{v'} dx \stackrel{(1)}{=} \frac{\sin x}{x^{1/4}} \Big|_1^{+\infty} + \frac{1}{4} \int_1^{+\infty} \frac{\sin x}{x^{5/4}} dx = \sin 1 + \frac{1}{4} J$$

Since $J \stackrel{(1/2)}{\text{is}} \text{absol conv}$, then $I_4 \stackrel{(1/2)}{\text{is}} \text{conv}$.

Note
This also can be done by Cauchy criterion or Abel's test.

$$5. I_5 = \int_1^{+\infty} \frac{|\cos x|}{x^{1/4}} dx.$$

nearby: we have

$$\frac{|\cos x|}{x^{1/4}} \geq \frac{\cos^2 x}{x^{1/4}} = \frac{1 + \cos(2x)}{2x^{1/4}} = \frac{1}{2x^{1/4}} + \frac{\cos 2x}{2x^{1/4}} \stackrel{(1)}{}$$

Since $\int_1^{+\infty} \frac{1}{2x^{1/4}} dx \stackrel{(1)}{\text{is}} \text{conv}$ & $\int_1^{+\infty} \frac{\cos 2x}{2x^{1/4}} dx \stackrel{(1)}{\text{is}} \text{div}$, then

I_5 is div.

EX III. Determine the following limits:

$$1. \lim_{n \rightarrow +\infty} \frac{1}{n} \sum_{k=1}^n \frac{(2n+5k)^2}{n^2} = \lim_{n \rightarrow +\infty} \frac{1}{n} \sum_{k=1}^n \left(2 + 5\frac{k}{n}\right)^2 \stackrel{(1)}{=} \lim_{n \rightarrow +\infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right)$$

where $f(x) = (2+5x)^2$ is cont, Hence integ on $[0,1]$.

$$\begin{aligned} \text{Thus } \lim_{n \rightarrow +\infty} \frac{1}{n} \sum_{k=1}^n \frac{(2n+5k)^2}{n^2} &= \int_0^1 (2+5x)^2 dx \stackrel{(1/2)}{=} \\ &= \int_0^1 (4 + 20x + 25x^2) dx = \left[4x + 10x^2 + \frac{25}{3}x^3 \right]_0^1 \\ &= 4 + 10 + \frac{25}{3} \stackrel{(1)}{=} \end{aligned}$$

$$2. \lim_{n \rightarrow +\infty} \frac{1}{n} \sum_{k=1}^n (\ln(5k+n) - \ln(n)) = \lim_{n \rightarrow +\infty} \frac{1}{n} \sum_{k=1}^n \ln\left(5\frac{k}{n} + 1\right) \stackrel{(1)}{=} \lim_{n \rightarrow +\infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right)$$

where $f(x) = \ln(5x+1)$ is cont, Hence integ on $[0,1]$.

$$\begin{aligned} \text{Thus } \lim_{n \rightarrow +\infty} \frac{1}{n} \sum_{k=1}^n (\ln(5k+n) - \ln(n)) &= \int_0^1 \ln(5x+1) dx \stackrel{(1/2)}{=} \quad (\text{By parts}) \\ &= x \ln(5x+1) \Big|_0^1 - \int_0^1 x \cdot \frac{5}{5x+1} = \ln 6 - \int_0^1 \frac{5x(1-1/5)}{5x+1} dx \\ &= \ln 6 - x \Big|_0^1 + \frac{1}{5} \ln|5x+1| \Big|_0^1 = (\ln 6) - 1 + \frac{1}{5} \ln 6 = \left(\frac{6}{5} \ln 6\right) - 1. \stackrel{(1)}{=} \end{aligned}$$

Ex III Study the nature of the following series.

(2) 1. $\sum_{n=1}^{+\infty} \frac{1}{n}$ div ($\because \frac{1}{n} \not\rightarrow 0$).

2. $\sum_{n=1}^{+\infty} \frac{n!}{n^n}$; $\left| \frac{u_{n+1}}{u_n} \right| = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = (n+1) \cdot \frac{n^n}{(n+1)^{n+1}}$
 $= \left(\frac{n}{n+1} \right)^n = e^{\ln \left(\frac{n}{n+1} \right)^n} = e^{n \ln \left(1 - \frac{1}{n+1} \right)}$
 $\underset{n \rightarrow +\infty}{\sim} e^{\frac{n \cdot (-1)}{n+1}} \rightarrow e^{-1} = \frac{1}{e} < 1.$

Since $\lim_{n \rightarrow +\infty} \left| \frac{u_{n+1}}{u_n} \right| = \frac{1}{e} < 1$, then by Ratio Test the series converges absol.

3. $\sum_{n=1}^{+\infty} \sin \left(\frac{n!}{n^n} \right).$

By 2., $\lim_{n \rightarrow +\infty} \frac{n!}{n^n} = 0.$

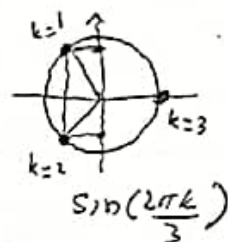
So $\sin \frac{n!}{n^n} \sim \frac{n!}{n^n} > 0$. using equivalent test & 2.

We obtain that the series conv.

4. $\sum_{n=2}^{+\infty} \frac{1}{\ln n} \sin \left(\frac{2\pi n}{3} \right)$; using Dirichlet's test

(1) $\frac{1}{\ln n}$ is positive & $\frac{1}{\ln n} \rightarrow 0$.

$\therefore B_n = \sum_{k=1}^n \sin \left(\frac{2\pi k}{3} \right)$ is bdd: ???



$B_n = \begin{cases} \frac{\sqrt{3}}{2} & \text{if } n=1 \\ 0 & \text{if } n=2 \\ 0 & \text{if } n=3 \end{cases} \Rightarrow B_n = \begin{cases} \frac{\sqrt{3}}{2} & \text{if } n=1+3m \\ 0 & \text{if } n=2+3m \\ 0 & \text{if } n=3+3m \end{cases} m \in \mathbb{N}^* \Rightarrow |B_n| \leq \frac{\sqrt{3}}{2}$

(2)

Thus the series is conv by Dirichlet test.

Ex IV. 5. $\sum_{n=1}^{+\infty} \frac{(-1)^n + 1}{\sqrt{n}}$

① $\sum_{n=1}^{+\infty} \frac{(-1)^n}{\sqrt{n}}$ is conv (by alternating series test $\{\frac{1}{\sqrt{n}}\} \searrow \rightarrow 0$)

① $\sum_{n=1}^{+\infty} \frac{1}{\sqrt{n}} = \sum_{n=1}^{+\infty} \frac{1}{n^{1/2}}$ div ($\alpha = 1/2 \leq 1$)

④ Hence the series diverges.

Ex V. Study according to the values of $\alpha \in \mathbb{R}$, the nature of the following series.

1. $I_\alpha = \int_0^{+\infty} \frac{\arctan x}{x^\alpha} dx$

near 0+: $\frac{\arctan x}{x^\alpha} \underset{0^+}{\sim} \frac{x}{x^\alpha} = \frac{1}{x^{\alpha-1}}$ conv $\Leftrightarrow \alpha-1 < 1 \Leftrightarrow \alpha < 2$

near $+\infty$: $\frac{\arctan x}{x^\alpha} \underset{+\infty}{\sim} \frac{\pi/2}{x^\alpha} > 0$ conv $\Leftrightarrow \alpha > 1$

Thus I_α conv iff $1 < \alpha < 2$.

2. $J_\alpha = \int_2^{+\infty} \frac{1}{(x-2)\sqrt{x-3}} dx$

① near $+\infty$: $f(x) \underset{+\infty}{\sim} \frac{1}{x\sqrt{x}} = \frac{1}{x^{3/2}} > 0$ conv

① near 3: $f(x) \underset{3}{\sim} \frac{1}{\sqrt{x-3}} = \frac{1}{|x-3|^{1/2}} > 0$ conv.

① near 2: $f(x) \underset{2^+}{\sim} \frac{1}{(x-2)^1} \left(\frac{+\infty}{\neq 0} \right)$ div

④ Hence J_α conv for $\alpha > 2$.

Bonus

3. $\sum_{n=1}^{+\infty} \ln(1+n^\alpha)$

- ① • $\alpha > 0 \Rightarrow n^\alpha \rightarrow +\infty \Rightarrow \ln(1+n^\alpha) \rightarrow +\infty \Rightarrow$ The series diverges
- ① • $\alpha = 0 \Rightarrow \ln(1+n^\alpha) = \ln 2 \not\rightarrow 0 \Rightarrow$ The series div.
- ① • $\alpha < 0 \Rightarrow n^\alpha \rightarrow 0 \Rightarrow \ln(1+n^\alpha) \sim n^\alpha = \frac{1}{n^{-\alpha}}$ (conv for $-\alpha > 1$)
i.e. $\alpha < -1$

→ Hence the series conv iff $\alpha < -1$.

Bonus ← $\left(\frac{1}{2}\right)$

4. $\sum_{n=1}^{+\infty} \frac{\alpha^n}{n!}$

$$\left| \frac{u_{n+1}}{u_n} \right| = |\alpha| \frac{n}{n+1} \rightarrow |\alpha|$$

- ① • If $|\alpha| > 1$, i.e. $(\alpha \in]-\infty, -1[\cup]1, +\infty[)$, then the series conv.
- ① • If $|\alpha| < 1$, i.e. $\alpha \in]-1, 1[$, then the series div.
- ① • If $\alpha = -1$: $\sum_{n=1}^{+\infty} (-1)^n \cdot \frac{1}{n!}$ conv (Alternating series $\left\{ \frac{1}{n!} \right\} \xrightarrow{n \rightarrow +\infty} 0$)

← Bonus ① • If $\alpha = 1$: The series becomes $\sum_{n=1}^{+\infty} \frac{1}{n!}$; $\frac{1}{n!} = \frac{1}{1 \cdot 2 \cdot \dots \cdot (n-1) \cdot n} \leq \frac{1}{n(n-1)} \sim \frac{1}{n^2} > 0$
hence by comp & equiv test, the series conv.

Then the series converges iff $\alpha \in]-\infty, -1] \cup [1, +\infty[$.

Ex IV. Let $m \in \mathbb{N}$, f cont near 0 s.t. $f(t) = o(t^m)$

1. $F(x) = \int_0^x f(t) dt$, s.t. $\lim_{x \rightarrow 0} \frac{|F(x)|}{|x|^{m+1}} = 0$ using M.I.V.T for integrals

$$F(x) = \int_0^x 1 \cdot f(t) dt \stackrel{(1)}{=} f(c_x) \int_0^x 1 \cdot dt \quad 0 < c_x < x \text{ or } x < c_x < 0$$

$$= x f(c_x)$$

$$\text{Now, } \frac{|F(x)|}{|x|^{m+1}} = \frac{|x| |f(c_x)|}{|x|^{m+1}} = \frac{|f(c_x)|}{|x|^m} \begin{cases} 0 < c_x < x \text{ or } x < c_x < 0 \\ \Rightarrow |c_x| < |x| \\ \Rightarrow |c_x|^m < |x|^m \end{cases}$$

$$\leq \frac{|f(c_x)|}{|c_x|^m} \stackrel{(1)}{\xrightarrow{x \rightarrow 0}} 0 \quad \left(\because \lim_{t \rightarrow 0} \frac{f(t)}{t^m} = 0 \right)$$

Hence $\lim_{x \rightarrow 0} \frac{|F(x)|}{|x|^{m+1}} = 0$

2. $\frac{|F(x)|}{|x|^{m+1}} \xrightarrow{x \rightarrow 0} 0 \stackrel{(1)}{\Rightarrow} \frac{F(x)}{|x|^{m+1}} \xrightarrow{x \rightarrow 0} 0 \Rightarrow F(x) = o(x^{m+1})$

3. $\underbrace{\frac{1}{1+t^2} - [1 - t^2 + t^4 - \dots + (-1)^m t^{2m}]}_{f(t)} = o(t^{2m+1})$

* f is cont near 0 (sum of cont fns).

$$\Rightarrow F(x) = \int_0^x f(t) dt = \text{Arctan } x - \left[x - \frac{x^3}{3} + \frac{x^5}{5} - \dots + (-1)^m \frac{x^{2m+1}}{2m+1} \right] = o(x^{2m+2})$$

$$\Rightarrow \text{Arctan } x \stackrel{(1)}{=} x - \frac{x^3}{3} + \frac{x^5}{5} - \dots + (-1)^m \frac{x^{2m+1}}{2m+1} + o(x^{2m+2})$$

Ex. VII. $f: [0,1] \rightarrow \mathbb{R}$ cont, $a_n = \frac{1}{n} \int_0^1 t^n f(t) dt$. S.T. $\sum_{n=1}^{\infty} a_n$ is abs conv.

f cont on $[0,1] \Rightarrow f$ bdd on $[0,1] \Rightarrow |f(x)| \leq M$ for some M . $\circledast$

$$\begin{aligned} \text{Hence } |a_n| &= \frac{1}{n} \left| \int_0^1 t^n f(t) dt \right| \leq \frac{1}{n} \int_0^1 t^n |f(t)| dt \\ &\leq \frac{1}{n} M \int_0^1 t^n dt \\ &\leq \frac{1}{n} M \left[\frac{t^{n+1}}{n+1} \right]_0^1 \\ &\leq \frac{M}{n(n+1)} \leq \frac{M}{n^2} \end{aligned}$$

Thus the series conv abs.